

Emanuele Bugliarello

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Research Interests

Multimodal and multilingual natural language processing, representation learning, transfer learning

Education

PhD in Computer Science 2019–present
University of Copenhagen Copenhagen, Denmark
Advisors: Desmond Elliott, Anders Søgaard

MSc in Communication Systems, Data Analytics specialisation 2015–2018
École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Thesis: *Matrix Completion in the Unit Hypercube via Structured Matrix Factorization*
GPA: 5.53/6 (3/44)

BEng in Electronics and Information Engineering 2013–2018
Tongji University (同济大学) Shanghai, China
Thesis: *General large-batch methods for scalable accurate training of neural networks*
GPA: 29.34/30

BSc in Telecommunications Engineering 2012–2015
Politecnico di Torino Turin, Italy
GPA: 28.86/30 (3/120) – Graduation Grade: 110/110 *Cum Laude*

Machine Learning Schools: LxMLS 2020, Oxml 2020, ALPS 2021

Experience

Research Scientist Intern with Lisa Anne Hendricks and Aida Nematzadeh 06/2022–present
DeepMind, Language Team London, UK

Visiting Researcher with Prof. Siva Reddy 01/2022–05/2022
Mila – Quebec AI Institute Montreal, Canada

Research Scientist Intern with Rishabh Mehrotra and Mounia Lalmas 08/2020–11/2020
Spotify Research, Tech Research Team London, UK
On Multi-Objective Music Sequencing via Counterfactual Beam Search

Research Associate with Prof. Naoaki Okazaki 10/2018–09/2019
Tokyo Institute of Technology (東京工業大学), Okazaki Lab Tokyo, Japan
On Neural Machine Translation: Dependency-aware Self-Attention and Cross-Mutual Information

Visiting Scholar with Prof. Rui Fan and Prof. Dan Alistarh 04/2018–07/2018
ShanghaiTech University (上海科技大学) Shanghai, China
Large-Batch Algorithms for Deep Learning at Scale

Research Intern with Swayambhoo Jain, Brian Eriksson and Jean Bolot 09/2017–03/2018
Technicolor AI Lab Palo Alto, CA, USA
Intelligent Visual Effects Scheduling via Constrained Matrix Factorization

Research Assistant with Marcel Schöngens, Maxime Martinasso and Claudio Gheller 06/2017–09/2017
Swiss National Supercomputing Centre (CSCS) Lugano, Switzerland
Distributed TensorFlow: A performance evaluation

Teaching

Teaching Assistant, EPFL: Internet Analytics (COM-308) Spring 2017
Teaching Assistant, UCPH: Advanced Topics in Natural Language Processing (NDAK19001U) Winter 2020
Teaching Assistant, UCPH: Neural Information Retrieval (NDAK20002U) Spring 2021

Selected Publications

7. **IGLUE: A Benchmark for Transfer Learning across Modalities, Tasks, and Languages**
E. Bugliarello, F. Liu, J. Pfeiffer, S. Reddy, D. Elliott, EM. Ponti, I. Vulić
In *ICML 2022*
6. **Visually Grounded Reasoning across Languages and Cultures**
F. Liu*, E. Bugliarello*, EM. Ponti, S. Reddy, N. Collier, D. Elliott
In *EMNLP 2021* **Best Long Paper Award**
5. **Vision-and-Language or Vision-for-Language? On Cross-Modal Influence in Multimodal Transformers**
S. Frank*, E. Bugliarello*, D. Elliott
In *EMNLP 2021*
4. **Multimodal Pretraining Unmasked: A Meta-Analysis and a Unified Framework of Vision-and-Language BERTs**
E. Bugliarello, R. Cotterell, N. Okazaki, D. Elliott
TACL 2021
3. **The Role of Syntactic Planning in Compositional Image Captioning**
E. Bugliarello, D. Elliott
In *EACL 2021*
2. **It's Easier to Translate *out of* English than *into* it: Measuring Neural Translation Difficulty by Cross-Mutual Information**
E. Bugliarello, SJ. Mielke, A. Anastasopoulos, R. Cotterell, N. Okazaki
In *ACL 2020*
1. **Enhancing Machine Translation with Dependency-Aware Self-Attention**
E. Bugliarello, N. Okazaki
In *ACL 2020*

Invited Talks

- **Towards Multilingual Vision-and-Language Models**
Heidelberg University (ICL) 06/2022
- **IGLUE: A Benchmark for Transfer Learning across Modalities, Tasks, and Languages**
Google Research 02/2022
University of Cambridge (LTL) 03/2022
- **Visually Grounded Reasoning across Languages and Cultures**
University of Trento (CIMEC) 06/2021
Samsung AI Center - Toronto 11/2021

Skills

Programming

- Python, Matlab
- NumPy, Pandas, scikit-learn
- PyTorch
- CUDA, SQL

Languages

- native: Italian
- fluent: English (TOEFL iBT 109)
- intermediate: French
- basic: Chinese (HSK III), Japanese, Danish

Fellowships, Honors, and Awards

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| EuroHPC Grant with Desmond Elliott on "Vision and Language on a Shoestring without Tear" | 06/2021 |
| University of Copenhagen Marie Skłodowska-Curie TALENT Doctoral Fellowship | 06/2019 |
| EPFL PhD Program in Computer Science Fellowship [offer declined] | 05/2019 |
| The Japanese Government (MEXT) Scholarship [offer declined] | 01/2019 |
| PoliTong Merit Scholarship | 07/2013 |
| MIUR National Register of Excellences | 06/2012 |

Service

- Reviewer:** ARR, EMNLP, ICLR, ICML, IJCAI, NeurIPS
Organiser: Multilingual Multimodal Learning Workshop (ACL 2022)